What Does Fine-Tuning Buy a Small Vision-Language Decision Model?

Abstract and summary of the paper

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Abstract

A decision model answers typed questions about a state made of text and images by returning a probability for every allowed option in one forward pass, without generating text. We study what fine-tuning a small vision-language model (Qwen3-VL-2B with LoRA and a learned option-scoring head) buys over the baselines a practitioner would try first, using an implementation that trains on an 8 GB laptop GPU. Measured against the base model with post-hoc temperature scaling, fine-tuning raises accuracy by 6.9 points (95% CI 5.5 to 8.3) on 16 held-out datasets, but by only 2.8 points (1.2 to 4.4) on 11 further datasets that influenced no design decision, and not significantly on three zero-shot sources. Temperature scaling alone removes 74 to 87% of the calibration error that separates the raw base model from the fine-tuned one, so most of the calibration benefit that is easy to attribute to fine-tuning is not due to it. Over three seeds, training on a broader mix of tasks beats a narrower mix on all five held-out collections (+1.3 to +5.1 points), whereas a learned answer head is no more accurate than a plain letter-logit readout, which generalises better to unseen sources but cannot handle more than 26 options. The fine-tuned model falls below the base model on questions with more than ten options; averaging over option orders recovers part of the gap. We report every comparison with confidence intervals and state the limits of the evidence.

The question, in plain language

Many systems must pick one answer from a fixed list and say how confident they are: routing a request, flagging a message, choosing a tool. We took a small publicly available AI model that reads text and images and fine-tuned it for exactly this job. The paper asks a plain question: **after all that work, how much better is it than the same untouched model with a simple, free adjustment of its confidence?** Earlier comparisons usually leave that adjustment out, which makes fine-tuning look better than it is.

Key findings

- **Fine-tuning helps, by less than it first appears.** About 7 points more accurate on 16 held-out datasets, but about 3 points on 11 further datasets that played no role in any decision, and no clear gain on three zero-shot sources.
- **Most of the confidence improvement is free.** A simple one-number adjustment (temperature scaling) applied to the untouched model removes 74 to 87% of the confidence error that fine-tuning is usually credited with fixing.
- **Training on more kinds of tasks was the one change that helped consistently**: +1.3 to +5.1 points over five collections, averaged over three training runs each.
- **A learned answer-scoring component is not more accurate than simply reading the model's letter probabilities.** It is better calibrated and the only option once a question has more than 26 choices.
- **A weakness remains.** On questions with more than ten options the fine-tuned model is below the untouched model; averaging over option orders recovers part of the gap, and padding the training data with extra options did not help.

Results at a glance

Table 1. Accuracy (%). The base model with temperature scaling is the strongest simple baseline; the last column is the released model minus that baseline, with a 95% interval.

Test collection	Base model + temp. scaling	Fine-tuned model	Difference
16 open datasets (3,678 questions)	71.1	78.0	+6.9 [+5.5, +8.3]
11 confirmation datasets (2,306 questions)	69.2	72.0	+2.8 [+1.2, +4.4]
2 fresh datasets (600 questions)	61.2	76.2	+15.0 [+11.2, +18.8]
3 zero-shot sources (720 questions)	68.3	70.0	+1.7 [-1.3, +4.3], not significant

The confirmation datasets were assembled after the comparison was finished and influenced no decision, so they are the best guide to performance on a new dataset.

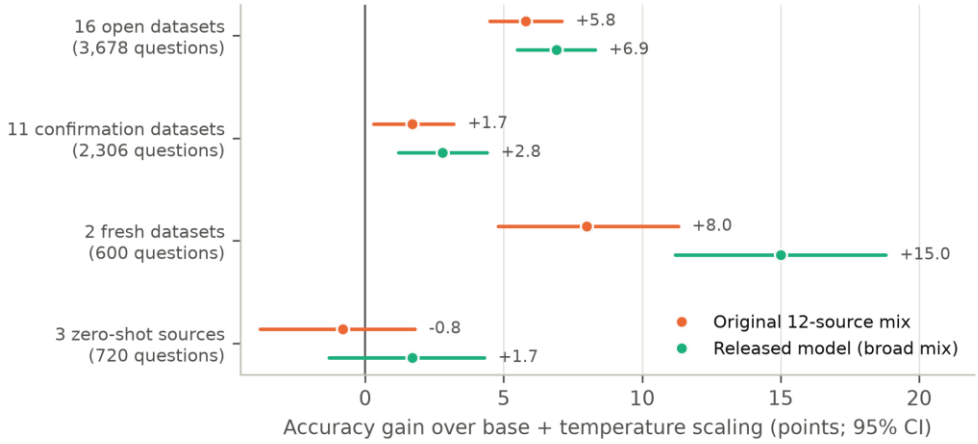


Figure 1. Accuracy gain over the baseline with 95% intervals. The gain is large on datasets close to the training families and small on the confirmation datasets.

How strong is the evidence?

- One base model, one laptop GPU and one training configuration; larger models or longer training could behave differently.
- Three training runs per recipe for the main comparison, but only one for some of the other techniques.
- No comparison against other released decision models, which would be the most informative next experiment.
- A leaderboard-style run was done only for an earlier version of the model and is unofficial; it should not be compared with published leaderboard scores.
- The experiments, analysis and drafts were produced with the assistance of an AI model working under the author's direction.

Publication fit

The paper is an empirical study, not a new method. It is suitable for a preprint, a workshop on calibration or small models, or a journal that judges the soundness of the evidence rather than novelty. It is unlikely to be accepted at the main track of a top conference. Before submission it needs the author's own review, a spot-check of the dataset citations, the institution's plagiarism check and, ideally, an outside reader.